

ALEXANDER PROSHKIN

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EDUCATION

UNIVERSITY OF CALIFORNIA, BERKELEY

EXPECTED GRADUATION: MAY 2027

B.S. in Electrical Engineering and Computer Sciences

GPA: 4.0/4.0

- **Relevant Coursework:** AI, ML, DUL, RL, Optimization, Comp Arch, C/C++, Java, Python, SQL, SWE, Cell Biology

PUBLICATIONS

IN-CONTEXT LEARNING WITH VISION TRANSFORMERS: CASE STUDY. [ARXIV:2505.20872](https://arxiv.org/abs/2505.20872)

MAY 2025

- Extending in-context learning to vision domain, demonstrating transformers can learn CNN and ViT functions from image prompts, achieving comparable performance to gradient-based training through novel curriculum approach.

WORK EXPERIENCE

SKEDAI

SAN DIEGO, CA

Co-Founder

May 2018 – Present

- Architecting constraint programming solution using **OR-Tools CP-SAT** solver to optimize multi-dimensional task scheduling with spatial, temporal, and resource constraints and hierarchical task creation with natural language processing
- Building distributed solver architecture in **TS** with central dispatcher to parallelize constraint solving across instances
- Designing **PostgreSQL** schema supporting objective weighting, multi-factor optimization, and real-time recalculation
- Leading technical recruitment and team management, conducting interviews and mentoring developers in sprints

ROBOT LEARNING LAB @ BAIR

BERKELEY, CA

Researcher

June 2025 – Present

- Developed information-theoretic exploration agents (MaxInfoRL) with ensemble disagreement under Pieter Abbeel for a 6-DoF **Agilex Piper** arm, scaling training on **Slurm** HPC clusters (**Berkeley** Research Computing, **ETH Zürich** Euler)
- Diagnosed and resolved curiosity traps and other negative emergent behaviors to enable intrinsic-reward-only grasping; completed graduate coursework in Deep Unsupervised Learning, Robotics, and Reinforcement Learning

CS 182, DATA 100 AT UC BERKELEY

BERKELEY, CA

UCS1 (Course Staff)

January 2025 – Present

- Leading office hours, discussion sections, and contributing to new curriculum development & **PrairieLearn** automation
- Deploying ACCESS GPU clusters with JupyterHub integration to provide students seamless access to shared compute

VICTOR CANERA DIAMONDS

BERKELEY, CA

Contract Software Developer

January 2025 – May 2025

- Integrated GIA API for dynamic product data, implemented fair pricing algorithms, vendor CSV ingestion, and price alert features; designed React frontend and **JS** backend to support vendor tagging, CPC tracking, and real-time analytics

NEXSTREAM TECHNICAL EDUCATION

SAN DIEGO, CA

Educational Software Developer

June 2023 – December 2023

- Researching trends in the EdTech sector to inform long-term business strategy and product development initiatives.
- Utilizing **nbgrader** to create autograded Python notebooks for teaching advanced K-12 students the fundamentals of ML

DRS DAYLIGHT SOLUTIONS

SAN DIEGO, CA

Group Lead

August 2022 – December 2022

- Engineering an IoT Ultra High Frequency RFID tag detection system with ESP-32 communication with **SQLServer**
- Designing a hidden markov model algorithm, enabling real-time item location tracking accounting for detection errors

GLOBAL HEALTHCARE EXCHANGE

LOUISVILLE, CO

Software Developer

June 2021 – July 2021

- Creating a web application in **HTML / CSS / JS** for UI identification of relevant information from non-standardized documents, converting the data into JSON corpus for integration into an AI model training pipeline

EXTRACURRICULAR EXPERIENCE

COURSEPATH INITIATIVE

SAN DIEGO, CA

Founder

August 2023 – April 2024

- Developing a Puppeteer-based scraper to extract transferable coursework, compiling results into a centralized platform.
- Collaborating with the dean to distribute the tool to students seeking guidance on formulating multiyear transfer plans

IEEE-ETA KAPPA NU EECS HONORS SOCIETY

BERKELEY, CA

DeCal Candidate / Computing Services Officer / Activities Officer

August 2024 - Present

- Organizing logistics for the course "Going Down the EECS Stack," coordinating with professors, students as instructors.
- Designing and developing website using **Django**, hosting tutoring sections for lower division EE/CS courses
- Organizing Retreat Planning + Activities for over 60 new members, building lasting community traditions + connections